ASSIGNMENT-2

(6) Derive the best, average and worst case time complexity for Quick Sort.

(A) The complexities of quicksort are as follows

(i) Best case (O(nlogn))

- It is when the array is divided exactly into two halves.

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

using masters theorem

$a = 2, b = 2 \quad f(n) = cn = \Theta(n)$

$$n^{\log_b^a} = n^{\log_2^2} = n$$

$\therefore f(n) = \Theta(n^{\log_b^a})$ -> case (2)

$\therefore T(n) = \Theta(n \log n)$

Best Time complexity = $\Theta(n \log n)$

(ii) Average case

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

$$T(n) = 2\left[2T\left(\frac{n}{4}\right) + c\frac{n}{2}\right] + cn$$

$$= 4T\left(\frac{n}{4}\right) + 2cn$$

$$= 8T\left(\frac{n}{8}\right) + 3cn$$

$$\Rightarrow T(n) = 2^k T\left(\frac{n}{2^k}\right) + kcn$$

$$\Rightarrow \frac{n}{2^k} = 1 \Rightarrow n = 2^k$$

$$\Rightarrow k = \log_2 n$$

$$\Rightarrow T(n) = nT(1) + cn \log_2 n$$

($\because T(1) = c$)

$$T(n) = cn + cn \log_2 n$$

$$\Rightarrow T(n) = O(n \log n)$$

($\because$ ignoring constants & low order terms)

(iii) Worst Case

. happens when the pivot is always the smallest or largest in the array.

$$\Rightarrow T(n) = T(n-1) + cn$$

$$T(n) = [T(n-2) + c(n-1)] + cn$$

$$= T(n-2) + c(n-1) + cn$$

$$\Rightarrow T(n-3) + c(n-2) + c(n-1) + cn$$

$$\Rightarrow T(n) = T(n-k) + c(n-k+1) + \cdots c(n-1) + cn$$

=> Rec stops when

$$n-k = 1$$

$$\Rightarrow \boxed{n-1 = k}$$

Sub $T(n) = T(1) + c(2+3+4 \cdots + n)$

$$= c + c(2+3+4+\cdots+n)$$

$$= \frac{n(n+1)}{2} - 1$$

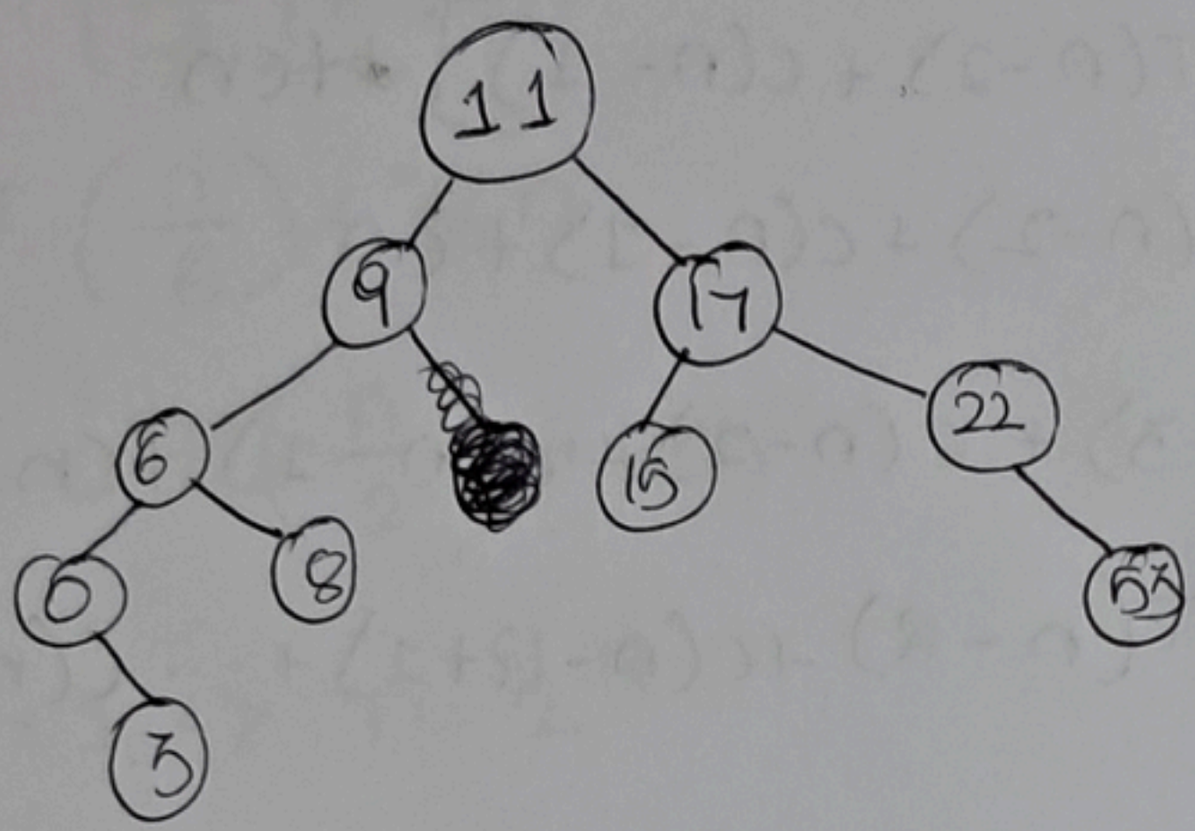
$$\Rightarrow T(n) = c + c\left(\frac{n(n+1)}{2} - 1\right)$$

$$= \frac{c}{2}(n^2+n)$$

$$\Rightarrow T(n) = O(n^2)$$

(7) Create a BST for 11, 9, 6, 17, 8, 22, 15, 6, 0, 63, 3 display BST, what is the height of BST? In which level Key 6 is present & Trace all the ancestors of 3.

(A)



(i) Height of BST is 5 considering root node

(ii) Key 6 is present in level 3.

(iii) ancestors of 3 are as follows.

$11 \rightarrow 9 \rightarrow 6 \rightarrow 0 \rightarrow 3$

(i) 4 edges are to be considered. Then the height is 4.

⑧ Hash function defined by $h(Key) = Key \bmod 6$, using Linear probing from 0 to 5 inserted (20, 10, 31, 19, 25, 28) loc of 19 & 25

0	28
1	31
2	20
3	19
4	10
5	25

final table

location of 19 is 4 at index 3 & 25 is 6 at index 5

$h(20) = 20 \% 6 = 2$

$h(10) = 10 \% 6 = 4$

$h(31) = 31 \% 6 = 1$

$h(19) = 19 \% 6 = 1$ (we will inbelt at 3)

$h(25) = 25 \% 6 = 1$

(inbelt a 5)

$h(28) = 28 \% 6 = 4$

(inbelt at 0)

(b) h(key) = key mod 7

20, 10, 31, 19, 25, 28

use both linear probing & seperate chaining

(i) Linear probing

20, 10, 31, 19, 25, 28

0	25
1	28
2	–
3	10
4	31
5	19
6	20

h(20) = 20 % 7 = 6

h(10) = 10 % 7 = 3

h(31) = 31 % 7 = 3

h(19) = 19 % 7 = 5

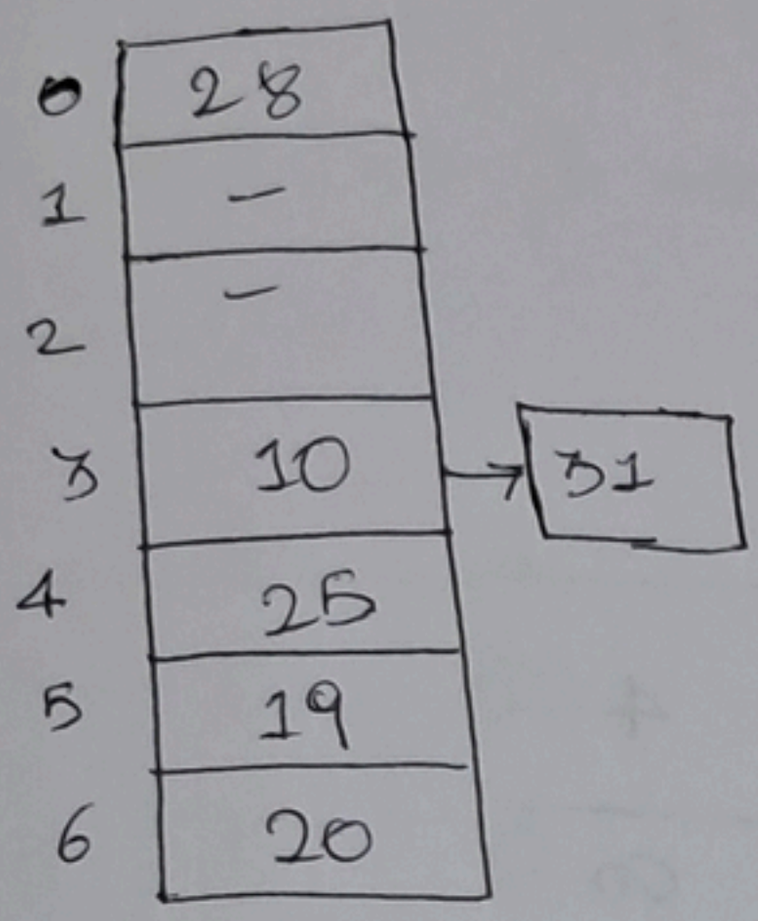
h(25) = 25 % 7 = 4

h(28) = 28 % 7 = 0

(ii) Separate chaining

20, 10, 31, 19, 25, 28

$h(12) = k \% 7$



$h(20) = 20 \% 7 = 6$

$h(10) = 10 \% 7 = 3$

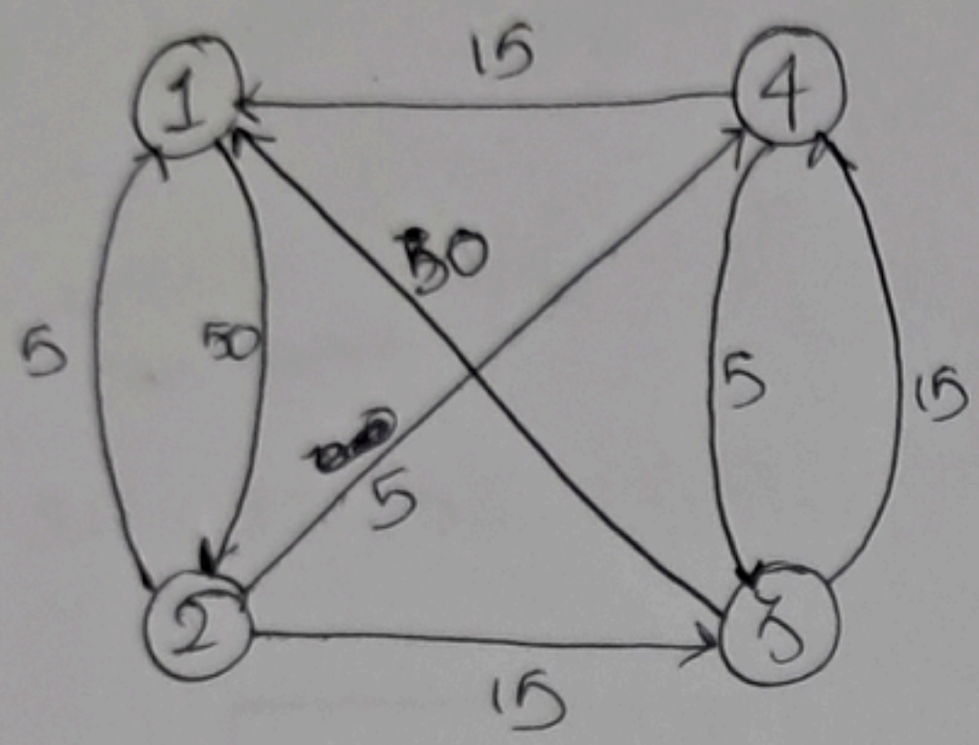
$h(31) = 31 \% 7 = 3$

$h(19) = 19 \% 7 = 5$

$h(25) = 25 \% 7 = 4$

$h(28) = 28 \% 7 = 0$

(8) Obtain Matrix A with shortest paths using all pairs shortest paths, for the given graph



$k=0$

	1	2	3	4
1	0	50	∞	∞
2	5	0	15	5
3	30	∞	0	15
4	15	∞	5	0

$k=1$

	1	2	3	4
1	0	50	∞	∞
2	5	0	10	5
3	30	80	0	15
4	15	65	5	0

$2 \to 3$ => the new path is $\times$

$2 \to 4 \to 3 = 5+5 = 10 < 15$ } $\times$

$4 \to 2 \Rightarrow 4 \to 1 \to 2$

$= 15 + 50 < \infty$

$3 \to 2 \Rightarrow 3 \to 1 \to 2 = 30 + 50 < \infty$

$= 80 < \infty$

$k = 2$

	1	2	3	4
1	0	50	65	55
2	5	0	~~15~~ 5	5
3	30	80	0	15
4	15	65	5	0

} only $1 \to 3$ & $1 \to 4$ are update

$k = 3$

	1	2	3	4
1	0	50	65	55
2	5	0	15	5
3	30	80	0	15
4	15	65	5	0

} no changes

$k = 4$

	1	2	3	4
1	0	50	60	55
2	5	0	10	5
3	30	80	0	15
4	15	65	5	0

} $1 \to 3$ & $2 \to 3$ are updated.